EIN 4343 Inventory and Supply Chain Systems
Spring 2026
Final Study Guidelines

Introduction, Deterministic Inventory Modeling, and Stochastic Inventory modeling sections are from the study guides provided previously. I marked off the topics that you should not worry about; there will be no T/F or essay type questions. Transportation, Logistics, and Supply Chain sections summarizes what you should be focusing on in the new material - "Sorry the Bullwhip Effect and Demand Forecasting; we are done…"

INTRODUCTION

- Reasons to keep inventory
- Costs associated with inventory
- ABC Analysis
    - Purpose
    - Procedure
    - Need to examine result to reclassify some items as necessary
- XYZ Analysis
    - Purpose
    - Procedure
- Four step approach to solving inventory problems (total cost per cycle, cycle length, etc…)

DETERMINISTIC INVENTORY MODELING

EOQ and its Extensions

- Assumptions of EOQ and their implications – I would not ask you to list modeling assumptions, but you should have a solid understanding of what the assumptions imply and how they facilitate model development.
- Derivation of single inventory cycle total cost expression – you should be able to replicate this based on the graphical representation of a single inventory cycle
- Derivation of inventory cycle length
- Derivation of average annual total cost function
- Derivation of EOQ expression – you should be able to replicate parts and entire derivation of the EOQ expression
- Be able to solve problems using EOQ expression to evaluate the optimal order quantity, inventory cycle length, average inventory level, number of orders per year as well as the average annual total cost
- Be able to derive the solutions and solve problems that involve extensions of EOQ under
    - Finite production rate
    - Finite constant lead time
    - Price discounts
        - All-units
        - Incremental
- ~~Be able to answer true or false, multiple-choice or short essay questions related to concepts~~

PRELIMINARIES: PROBABILTY REVIEW

- Review the properties of probability distribution/mass functions and the corresponding cumulative distribution functions
- Review the properties of expected value functions
- Given probability distribution function parameters for the demand and the lead time, you should be able to obtain the parameters for the demand during lead time
- Be very comfortable working with the z-table, reading z, $F(z)$ and $L(z)$ values and interpolating values as necessary

STOCHASTIC INVENTORY MODELING

Newsvendor Model and Its Extensions

- Assumptions of the newsvendor model and their implications – I would not ask you to list modeling assumptions, but you should have a solid understanding of what the assumptions imply and how they facilitate model development.
- Derivation of single-period cost minimization model and its optimization – you should be able to rework the derivation and apply Leibnitz's rule to obtain the result
- Derivation of single-period profit maximization model and its optimization
- Derivation of multiple period profit maximization model and its optimization under (i) full backlogging or (ii) lost sales
- Evaluation of order-up-to level q for any of the model variants above
- ~~Be able to answer true or false, multiple-choice or short essay questions related to concepts~~
- Be able to make simple proofs, e.g., "*Consider the multi-period newsvendor problem under full backlogging. Show that the total profit throughout the planning horizon is given by* $p(T-1)\mu$*"* Since it says "show that" you must make a mathematical argument not provide a description

in plain English. For this you would write down the expected sales in each period throughout the planning horizon and manipulate the expression to obtain the desired result.

(q,r) policy

- Assumptions for (q, r) policy and their implications – I would not ask you to list modeling assumptions, but you should have a solid understanding of what the assumptions imply and how they facilitate model development.
- Definition of safety stock level for model development
- Derivation of $n(r)$ and s under
  (i) full backlogging or
  (ii) lost sales
- Derivation of optimal policy parameters under
  (i) full backlogging or
  (ii) lost sales
- Evaluation of optimal policy parameters for a (q, r) policy under
  (i) full backlogging or
  (ii) lost sales using the iterative approach with the correct set of formulas for the order quantity and the reorder level
- Evaluation of the safety stock level for a (q, r) policy
- Evaluation of the expected total cost for a (q, r) policy

Service Level Considerations

- ~~Why are service levels needed?~~
- ~~Definitions of Type I and Type II service levels – I would not ask you to define it as an essay but you should be able to explain the difference between the two~~
- Identify parameters of a service level for a given Type I or Type II service level
- Identify implied Type I or Type II service level for a given policy

TRANSPORTATION

- A general awareness of the role of transportation decisions and their relationship to inventory decisions.

LOGISTICS

- You should know that there can be multiple elements of total logistics costs:
  - How is annual in-transit inventory holding cost calculated?
  - What is annual inventory holding cost? How is it calculated?
  - Why do annual inventory holding and transportation costs depend on each other?
- Be able to calculate the total cost for a given network configuration that use
  - Direct shipment using full truckloads
  - Direct shipment using less than full truckloads (i.e., optimal shipment sizes)
  - Shipments through a warehouse
  - Milk-run shipments
  - Etc.

SUPPLY CHAIN COORDINATION

- Be able to calculate replenishment order size and calculate associated total costs for each agent for a supply chain with two agents where either one of the agent has the channel power or when the two agents integrate their decision making